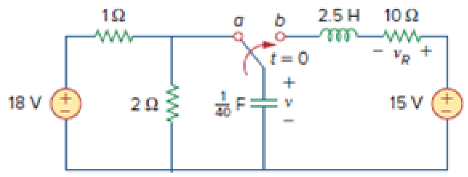
Chapter 8, Problem 13PP

Problem

Refer to the circuit in Fig. 8.21 (see Practice Prob. 8.7). Use *PSpice* to obtain *v*(*t*) for 0<*t*<2.

Fig. 8.21:



Step-by-step solution

Step 1 of 11

Refer to Figure 8.21 in the textbook.

Draw the modified circuit for $t<0$.

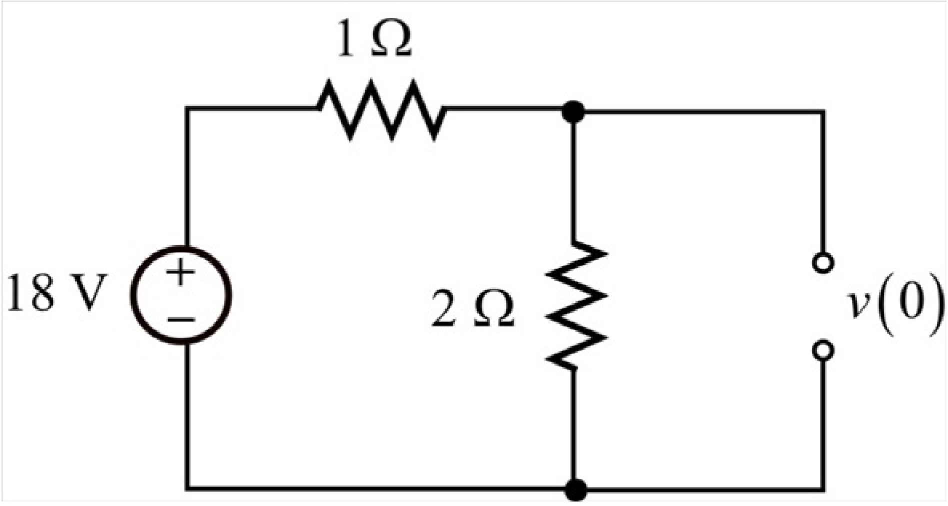


Figure 1

Step 2 of 11

Apply voltage division rule to calculate the initial voltage, $v(0)$.

$$\begin{aligned} v(0) &= \left[\frac{2}{2+1}\right](18) \\ &= \left[\frac{2}{3}\right](18) \\ &= 12\text{ V} \end{aligned}$$

The inductor current is independent of the circuit.

$$i(0)=0\text{ A}$$

Step 3 of 11

Draw the modified circuit for $t > 0$.

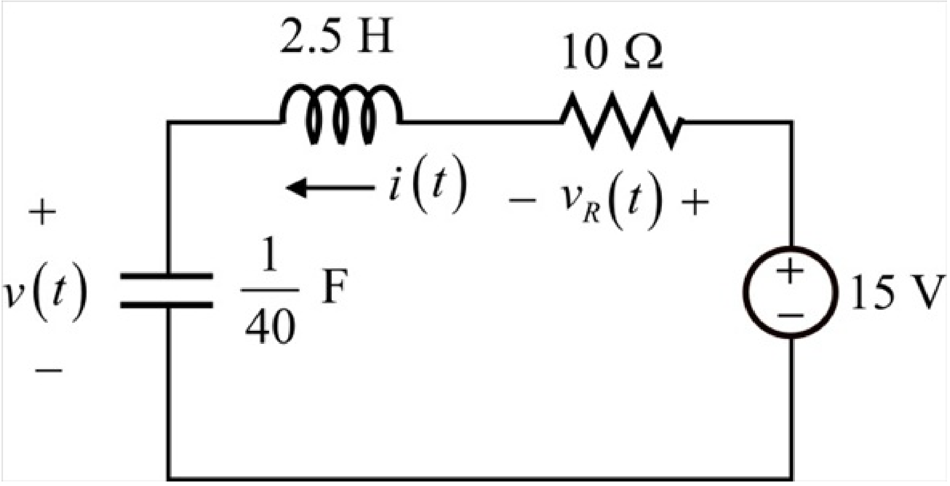


Figure 2

Step 4 of 11

Calculate the value of α.

$$\alpha = \frac{R}{2L}$$

Substitute $10\ \Omega$ for R and $2.5\ \text{H}$ for L.

$$\begin{aligned}\alpha &= \frac{10}{2(2.5)} \\ &= \frac{10}{5} \\ &= 2\end{aligned}$$

Calculate the value of ω_0.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute $\frac{1}{40}\ \text{F}$ for C and $2.5\ \text{H}$ for L.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{\frac{2.5}{40}}} \\ &= \sqrt{16} \\ &= 4\end{aligned}$$

Step 5 of 11

Since $\omega_0 > \alpha$, the response is under damped.

Calculate the characteristics of the system.

$$\begin{aligned}s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \\ &= -2 \pm \sqrt{4 - 16} \\ &= -2 \pm j3.464\end{aligned}$$

Calculate the total response.

$$v(t) = v_{ss} + \left(A_1 \cos \omega_d t + A_2 \sin \omega_d t\right) e^{-2t}$$

Here, v_{ss} is the steady state response. It is the final value of the capacitor voltage.

From the circuit, as $t \to \infty$, circuit reaches steady state and capacitor is replaced with open circuit and inductor with short circuit.

Draw the modified circuit with for $t \to \infty$.

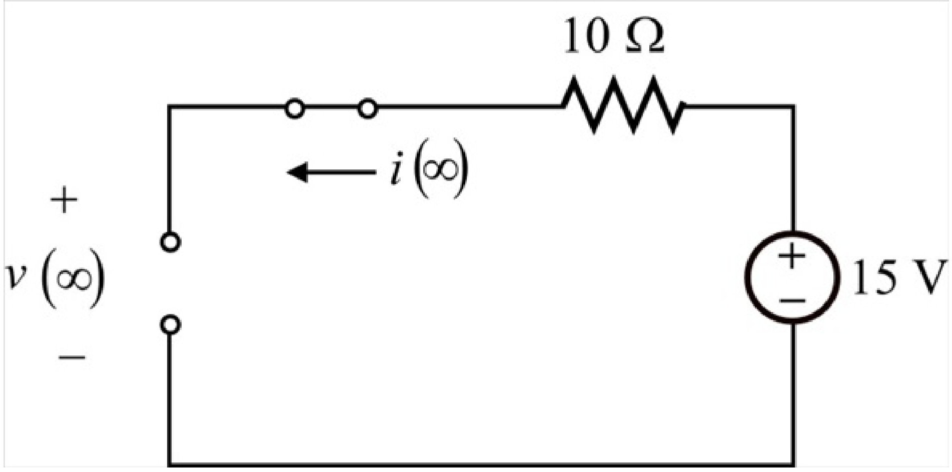


Figure 3

The steady state value is,

$$\begin{aligned} v_{ss} &= v(\infty) \\ &= 15 \text{ V} \end{aligned}$$

Substitute 15 V for v_{ss} in $v(t)$.

$$v(t) = 15 + (A_1 \cos 3.464t + A_2 \sin 3.464t)e^{-2t}$$

Substitute 0 for t.

$$\begin{aligned} v(0) &= 15 + (A_1 \cos 3.464(0) + A_2 \sin 3.464(0)) \\ 12 &= 15 + A_1 \\ A_1 &= -3 \end{aligned}$$

At $t = 0^+$, the inductor and capacitor are in series, so the current passing through it is same.

$$i(0) = C\frac{dv(0)}{dt}$$

$$\frac{dv(0)}{dt} = \frac{i(0)}{C}$$

$$\frac{dv(0)}{dt} = \frac{0}{\left(\frac{1}{40}\right)}$$

$$\frac{dv(0)}{dt} = 0 \text{ V/s}$$

Differentiate the function $v(t)$ with respect to t.

$$\begin{aligned} \frac{dv(t)}{dt} &= \frac{d}{dt}\left[15 + (A_1 \cos 3.464t + A_2 \sin 3.464t)e^{-2t}\right] \\ &= \begin{bmatrix} -2e^{-2t}(A_1 \cos 3.464t + A_2 \sin 3.464t) \\ +e^{-2t}(-3.464 A_1 \sin 3.464t + 3.464 A_2 \cos 3.464t) \end{bmatrix} \end{aligned}$$

Step 9 of 11

Substitute 0 for t.

$$\frac{dv(0)}{dt} = -2A_1 + 3.464A_2$$

Substitute 0 for $\frac{dv(0)}{dt}$ and -3 for A_1.

$$0 = -2(-3) + 3.464A_2$$
$$3.464A_2 = -6$$
$$A_2 = -1.7321$$

Substitute -1.7321 for A_2 and -3 for A_1 in $v(t)$.

$$\begin{aligned} v(t) &= 15 + e^{-2t}\left[-3\cos(3.464t) - 1.7321\sin(3.464t)\right] \text{ V} \\ &= 15 - e^{-2t}\left[3\cos(3.464t) + 1.7321\sin(3.464t)\right] \text{ V} \end{aligned}$$

Hence, the voltage across the capacitor is,

$$\boxed{15 - e^{-2t}\left[3\cos(3.464t) + 1.7321\sin(3.464t)\right] \text{ V}}$$

Step 10 of 11

Draw the circuit in Pspice.

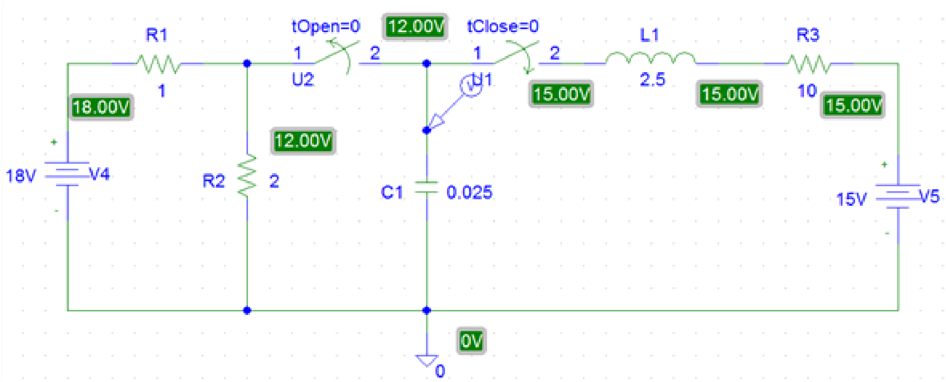


Figure 4

Step 11 of 11

Draw the following output:

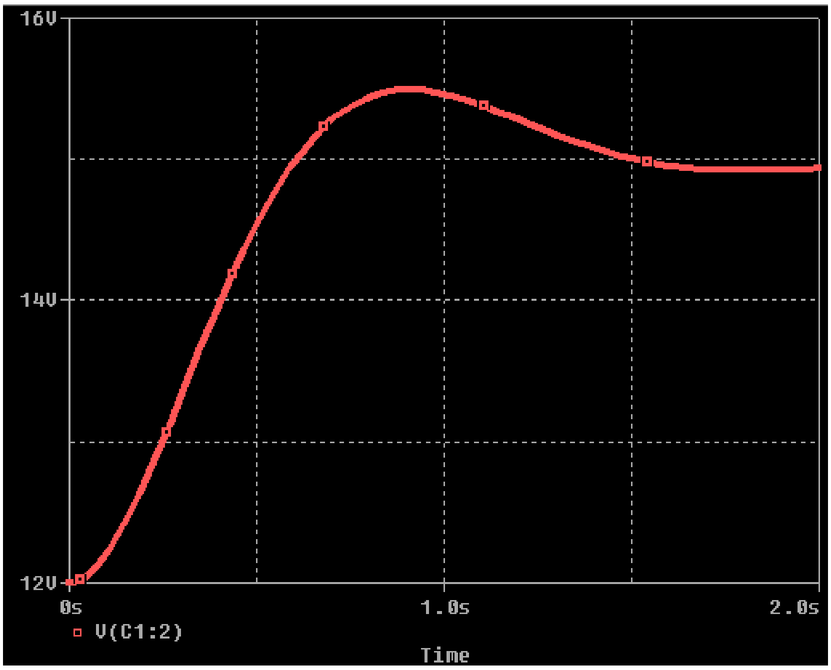


Figure 5